MATCH
Communications in Mathematical and in Computer Chemistry

MATCH Commun. Math. Comput. Chem. **00** (0000) 1–24
ISSN: 0340–6253
doi: 10.46793/match.00-00.xxxxx

The Loyola Index Family and the Resolution Limit of Degree-Ratio Indices on Trees

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(Received October 17, 2026)

Abstract

Every vertex-degree-based index that sees an edge only through the ratio of its end-degrees, such as the geometric–arithmetic, symmetric division deg and degree-ratio Sombor indices, depends on a graph only through its histogram of normalised edge imbalances $q_{uv} = |d_u - d_v|/(d_u + d_v)$, a direct consequence of Rada's structure relation. By exhaustive enumeration we determine when this histogram loses degree-pair information: it determines the degree-pair counts of a tree up to order twelve, and of a chemical tree up to order fifteen, but not from order thirteen (sixteen for chemical trees), where explicit colliding trees exist at every order. These are the first orders at which all degree-ratio indices fail simultaneously; a single index, such as the symmetric division deg index, can fail earlier. As a test case we use the *Loyola index family*

$$LO(G;\alpha,\beta,\gamma) \;=\; \sum_{uv\in E(G)} (d_u d_v)^{\alpha}\,(d_u + d_v)^{\beta}\, e^{\gamma q_{uv}},$$

an exponential extension of the Gutman–Milovanović family. By Rada's Lindemann–Weierstrass theorem its pure-imbalance points at algebraic $\gamma \neq 0$ separate exactly the imbalance histograms, and

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two chemical graphs exchange order at most five times along γ. Preregistered tests on measured boiling points did not demonstrate a predictive gain from γ; on those data ridge regression on the degree-pair counts had a higher cross-validated Q^2 than every single index evaluated, in point estimate. In an exploratory comparison, a gain on vaporisation enthalpies seen in a narrow search box shrinks to about zero once the parent is searched over a box that does not truncate it.

1 Introduction

Degree-based topological indices run from the Randić and Zagreb indices [20, 36] and their parametric extensions [7, 49] to the Sombor index [14] and its many variants [15]. Extremal results on bond-incident-degree (BID) indices are surveyed in [2]. On graphs with $\Delta \leq 4$ every such index is a linear functional of the ten edge-degree-pair counts [34], so no new index can carry more graph information than those counts. The question for a new index is therefore which count vectors it separates, and whether it predicts better than the counts themselves.

A growing group of indices sees each edge only through the ratio of its end-degrees: the geometric–arithmetic and arithmetic–geometric indices, the symmetric division deg index [45], the diminished Sombor index [29] and the degree-ratio indices [16, 32], including the degree-ratio Sombor index [4,18,40]. We show that all of them depend on a graph only through its histogram of normalised imbalances $q_{uv} = |d_u - d_v|/(d_u + d_v)$, and we determine when this histogram loses degree-pair information: for trees from order 13, and from order 16 for chemical trees.

We study this limit through the three-parameter extension

$$LO(G;\alpha,\beta,\gamma) = \sum_{uv\in E(G)} (d_u d_v)^{\alpha}(d_u + d_v)^{\beta} e^{\gamma q_{uv}}$$

of the Gutman–Milovanović family $M_{\alpha,\beta}$ [17,28], which we call the *Loyola index family*. It is a subfamily of the exponential vertex-degree-based indices of Rada [33], and its pure-γ points $LO(G;0,0,\gamma) = \sum_{uv} e^{\gamma q_{uv}}$ are the exponential indices of the edge function γq_{uv}. As $\gamma \to 0$, $LO(G;0,0,\gamma) =$

$m+\frac{\gamma}{2}\,IRLA(G)+O(\gamma^2)$, where $IRLA(G)=2\sum_{uv}q_{uv}$ [38] is a normalised form of Albertson's irregularity [1, 25]. Its extremal theory is developed in [8, 13, 21, 42], and parameter-optimised versions of the parent have already been fitted to octane properties [3]. The variable Euler–Sombor index [19] and the cosine-rule Sombor indices [31] weight the imbalance polynomially and without normalisation. The family is not proposed as a better predictor. Gutman's survey [15] observes that new variants hardly ever improve on earlier ones in applications, and our tests agree.

Our contributions are as follows.

(i) The ratio-plane theorem (Theorem 3): on the plane $\beta=-2\alpha$ the index depends only on the q-histogram, the pure-γ points at algebraic $\gamma\neq 0$ separate exactly the q-histograms, and distinct degree-pair profiles with equal histograms occur among trees of every order from 13 (from 16 for $\Delta\leq 4$). That these orders are the first is established by exhaustive enumeration. The corollary extends the collisions to every degree-ratio index.

(ii) At rational parent exponents a nonzero algebraic γ never destroys a distinction made by the parent (Proposition 4).

(iii) A bound on the order exchanges along γ (Proposition 6) and per-graph identifiability (Proposition 7). The extremal trees of the pure-γ points (Proposition 8) follow from known criteria and are included with a short proof.

(iv) Freeing γ gives no demonstrated gain once the parent's search box does not truncate it, and on measured boiling points ridge regression on the degree-pair counts leads every single index evaluated in point estimate (Section 5). On the octanes it does not.

2 Preliminaries

Let $G=(V,E)$ be a finite simple connected graph with at least one edge, of order n and size m, with degrees d_v, maximum degree Δ and minimum degree δ. The chemometric section uses hydrogen-suppressed alkane skeletons, for which $\Delta\leq 4$; the mathematical results do not assume this bound

unless stated. For integers $1 \le i \le j$ let $m_{ij}(G)$ be the number of edges with unordered degree pair (i,j); the vector $(m_{ij}(G))_{i\le j}$ is the *degree-pair profile* of G.

Definition 1 (Loyola index family)**.** For $\alpha, \beta, \gamma \in \mathbb{R}$,

$$LO(G;\alpha,\beta,\gamma) \;=\; \sum_{uv\in E(G)} (d_u d_v)^{\alpha}\,(d_u+d_v)^{\beta}\,\exp\!\left(\gamma\,\frac{|d_u-d_v|}{d_u+d_v}\right). \tag{1}$$

Write $q_{uv} = |d_u - d_v|/(d_u + d_v)$ and $q_{ij} = (j-i)/(i+j)$ for $i \le j$. Each summand depends only on the degree pair, so

$$LO(G;\alpha,\beta,\gamma) = \sum_{i\le j} m_{ij}(G)\,(ij)^{\alpha}(i+j)^{\beta}e^{\gamma q_{ij}}. \tag{2}$$

In particular $LO(G;\alpha,\beta,0) = M_{\alpha,\beta}(G)$, and γ has no effect on regular graphs. Special cases, catalogued in [28], include $M_1 = LO(0,1,0)$, $M_2 = LO(1,0,0)$, $HM = LO(0,2,0)$ [41], ${}^{m}M_2 = LO(-1,0,0)$, the Randić index $R = LO(-\frac{1}{2},0,0)$, the sum-connectivity index $\chi = LO(0,-\frac{1}{2},0)$ [48], $H/2 = LO(0,-1,0)$, $ISI = LO(1,-1,0)$, $GA = 2\,LO(\frac{1}{2},-1,0)$ [44] and $AG = \frac{1}{2}LO(-\frac{1}{2},1,0)$.

Definition 2 (q-histogram)**.** The q-histogram of G is $n_q(G) = \sum_{i\le j,\; q_{ij}=q} m_{ij}(G)$, $q \in [0,1)$.

Equal profiles give equal q-histograms; the converse fails (Theorem 3(iii)). Since $4ij/(i+j)^2 = 1 - q_{ij}^2$,

$$LO(G;\alpha,\beta,\gamma) = 4^{-\alpha} \sum_{uv\in E(G)} (d_u+d_v)^{2\alpha+\beta}\,(1-q_{uv}^2)^{\alpha}\,e^{\gamma q_{uv}}, \tag{3}$$

so $\kappa = 2\alpha + \beta$ acts only on the degree sum, and α, γ act only on q_{uv}, a function of the degree ratio.

Definition 3 (Ratio plane)**.** The ratio plane is $\Pi = \{(\alpha,\beta,\gamma) : \beta = -2\alpha\}$. It contains the pure-$\gamma$ points $(0,0,\gamma)$, GA and AG.

3 The resolution limit

3.1 The degree-pair ceiling

No vertex-degree-based index separates graphs with equal profiles, so the number of distinct profiles is a ceiling on distinct values.

Proposition 1 (Degree-pair ceiling)**.** *If $m_{ij}(G) = m_{ij}(H)$ for all $i \le j$, then $LO(G;\theta) = LO(H;\theta)$ for every $\theta \in \mathbb{R}^3$, and $TI(G) = TI(H)$ for every vertex-degree-based index TI. Among the* 18 *octane isomers exactly* 16 *distinct count vectors occur, and the pairs {3-methylheptane, 4-methylheptane} and {3,4-dimethylhexane, 3-ethyl-2-methylpentane} share their count vectors but differ in measured properties (Table 1). Hence every vertex-degree-based descriptor, tuned or not, carries a nonzero irreducible in-sample error on this benchmark.*

Proof. By (2), $LO(G;\theta)$ depends on G only through $(m_{ij}(G))_{i\le j}$, and so does every vertex-degree-based index [34], since these graphs share the M-polynomial [12]. The octane counts are computed and cross-checked in the reproducibility package [30]. ■

For the octane boiling points the resulting floor on the sum of squared errors of any fixed function of the counts is $2.925\,^{\circ}\mathrm{C}^2$, a root mean square of $0.403\,^{\circ}\mathrm{C}$. It bounds a single fixed predictor, not the cross-validated errors of Section 5, where each fold fits its own.

Table 1. The two degree-pair collision pairs among the 18 octane isomers, indistinguishable by every vertex-degree-based index. Units: T_B in °C, ΔH_f and ΔH_{vap} in $\mathrm{kcal\,mol^{-1}}$, S in $\mathrm{cal\,mol^{-1}\,K^{-1}}$; ω is dimensionless.

Isomer	T_B	ΔH_f	ΔH_{vap}	S	ω
shared count vector $2(1,2)$, $1(1,3)$, $2(2,2)$, $2(2,3)$:					
3-methylheptane	118.9	−50.82	9.52	111.26	0.371
4-methylheptane	117.7	−50.69	9.48	109.32	0.372
shared count vector $2(1,2)$, $2(1,3)$, $2(2,3)$, $1(3,3)$:					
3,4-dimethylhexane	117.7	−50.91	9.32	106.59	0.340
3-ethyl-2-methylpentane	115.6	−50.48	9.21	106.06	0.330

The ceiling is attained at almost every parameter point.

Proposition 2 (Generic discrimination)**.** *Let $m \neq m'$ be two profiles supported on the pairs $P_\Delta = \{(i,j) : 1 \le i \le j \le \Delta\}$, and write $z_{ij} = (\ln(ij), \ln(i+j), q_{ij})$. Then $D(\theta) = \sum_{(i,j)\in P_\Delta} (m_{ij} - m'_{ij})\, e^{\theta^\top z_{ij}}$ is a nonzero real-analytic function, whose zero set has Lebesgue measure zero in $\mathbb{R}^3$ [27]. Hence graphs with distinct profiles are separated by $LO(\cdot\,;\theta)$ for almost every θ.*

Proof. The map $(i,j) \mapsto z_{ij}$ is injective, since ij and $i+j$ determine the pair. Choose v with the $v^\top z_{ij}$ pairwise distinct, which excludes only finitely many hyperplanes of v. Along $\theta = tv$, D is an exponential sum with distinct exponents and coefficients not all zero, so it is not identically zero. ∎

The proof uses only the injectivity of $(i,j) \mapsto (\ln(ij), \ln(i+j))$, so it applies to the parent on the (α,β)-plane: generic discrimination is inherited, not conferred by γ. As there are countably many pairs of profiles, almost every point of the family is weakly discriminating, in the sense of Rada, Rodríguez and Sigarreta [35], over every class at once.

3.2 The ratio plane

Rada [34] shows that no vertex-degree-based index separates trees with equal profiles. The theorem below concerns a coarser loss: on Π, distinct profiles with equal q-histograms receive equal values.

Theorem 3 (Ratio plane)**.** *Let G and H be graphs.*

(i) On Π, $LO(G;\alpha,-2\alpha,\gamma) = \sum_q n_q(G)\,\big((1-q^2)/4\big)^\alpha e^{\gamma q}$ depends on G only through its q-histogram, and G and H agree at every point of Π if and only if $n_q(G) = n_q(H)$ for all q. Otherwise their values differ at every point of Π outside a set of measure zero.

(ii) For every nonzero algebraic γ, $LO(G;0,0,\gamma) = LO(H;0,0,\gamma)$ if and only if $n_q(G) = n_q(H)$ for all q. On any finite set of graphs the same holds for all but finitely many real γ.

(iii) Let G and H be trees with distinct edge-degree-pair profiles and equal q-histograms, each having a leaf adjacent to a vertex of the same

degree $d \geq 2$. Attaching a pendant path of length $L \geq 1$ at such a leaf in each produces trees with the same two properties and the same maximum degree. The profiles change by the same vector, whether or not the chosen leaf edge was the only $(1,d)$-edge. Consequently, for every $n \geq 13$ (and every $n \geq 16$ among trees with $\Delta \leq 4$) there are non-isomorphic trees of order n with distinct degree-pair profiles that no point of Π separates.

Proof. (i) The formula follows from (3) with $\beta = -2\alpha$. If the histograms differ, the difference at $\alpha = 0$ is $\sum_q (n_q(G) - n_q(H))\, e^{\gamma q}$, an exponential sum with distinct exponents and not all coefficients zero, which is not identically zero because the term with the largest exponent dominates as $\gamma \to +\infty$. The difference on Π is therefore a nonzero real-analytic function of (α, γ), whose zero set has measure zero [27]. (ii) Every imbalance is rational, so for algebraic $\gamma \neq 0$ the exponents γq are distinct algebraic numbers. By the Lindemann–Weierstrass theorem [5] the $e^{\gamma q}$ are linearly independent over the algebraic numbers, so the integer coefficients $n_q(G) - n_q(H)$ vanish. For real γ, a nonzero exponential sum with k distinct exponents has at most $k-1$ real zeros (Proposition 6). (iii) The attachment, a device of Rada [34], changes each profile by $-(1,d) + (2,d) + (L-1)(2,2) + (1,2)$, so distinct profiles stay distinct and equal histograms stay equal; the new vertices have degree at most 2. It remains to give the starting pairs. At order 13: T_1 is a path $c\,a\,b\,d\,e\,\ell$ in which c carries five pendant leaves, b and d one each and ℓ is a leaf, with edges $(1,6)^5, (2,6), (2,3)^2, (3,3), (1,3)^2, (1,2)$; T_2 is a path $c\,b\,x\,y\,z$ in which c carries five pendant leaves, b one and z two, with edges $(1,6)^5, (3,6), (1,3)^3, (2,3)^2, (2,2)$. Both have the q-histogram $\{0^1, \tfrac{1}{5}^2, \tfrac{1}{3}^1, \tfrac{1}{2}^3, \tfrac{5}{7}^5\}$ and a leaf adjacent to a vertex of degree 3. At order 16 with $\Delta \leq 4$: the profile $\{(1,2)^2, (1,4)^8, (2,4)^2, (4,4)^3\}$ is realised by a path $A\,B\,C\,D$ of vertices of degree 4, where A and D carry three pendant leaves each and B and C one pendant leaf and one pendant path of length 2 each; the profile $\{(1,4)^8, (2,2)^3, (2,4)^4\}$ is realised by vertices A, B, C of degree 4 joined as $A\,w\,B$ and $B\,v_1v_2v_3v_4\,C$ through vertices of degree 2, with three, two and three pendant leaves. Both have the q-histogram $\{0^3, \tfrac{1}{3}^4, \tfrac{3}{5}^8\}$ and a leaf adjacent to a vertex of degree 4. ∎

Part (ii) is Theorem 3.4 of [33] applied to the algebraic edge weights γq_{ij}, whose classes of equal weight are the q-classes. The construction in part (iii) parallels Example 2.2 of [35], where indices that depend only on the degree sum are defeated by chemical trees whose counts agree on classes of equal sum; here the classes are those of equal ratio, which for $\Delta \leq 4$ are $\{(1,2),(2,4)\}$ and $\{(2,2),(3,3),(4,4)\}$ (ignoring $(1,1)$, which occurs only in K_2). Song and Huang [43] determined the trees of order at most 10 with equal edge division vectors, which receive equal values of every edge-additive eccentric index; neither that partition of trees nor ours refines the other. We found no earlier determination of the orders 13 and 16, or of explicit colliding trees for degree-ratio indices, in a literature search recorded in the repository [30].

Corollary (Degree-ratio indices)**.** *Let* $\rho_{uv} = \min(d_u, d_v)/\max(d_u, d_v)$ *and* $I_f(G) = \sum_{uv \in E(G)} f(\rho_{uv})$ *for any function* f *on* $(0,1]$*. Since* $\rho_{uv} = (1-q_{uv})/(1+q_{uv})$*,* I_f *depends on* G *only through its* q*-histogram, and the trees of Theorem 3(iii) are not separated by any such index. This applies to GA, AG, the exponential geometric–arithmetic index [10], the diminished Sombor index [29], the symmetric division deg index [45], the inverse symmetric division deg index and its exponential [11], and the degree-ratio indices [16, 32], including the degree-ratio Sombor index [4, 18, 40].*

The corollary is the structure observation of [33, Proposition 2.5] for edge weights that depend only on the degree ratio; what is new are the explicit witnesses and the orders at which they first occur. Within this class the pure-γ points at algebraic $\gamma \neq 0$ attain the most possible, by part (ii). GA does not (Table 2). The parent attains it as well: at $\gamma = 0$ part (i) gives an exponential sum in α with distinct bases, which separates two distinct histograms at all but finitely many α. The limit thus belongs to the dependence on the degree ratio, not to γ. The orders 13 and 16 are the first at which the q-histogram, and hence every degree-ratio index simultaneously, fails to separate distinct profiles. A single index can fail earlier through coincidences among its values. The symmetric division deg index merges two distinct profiles of chemical trees already at order 9, for instance $\{(1,2),(1,3)^2,(2,2)^4,(2,3)\}$ and $\{(1,2)^3,(1,3),(2,3)^3,(3,3)\}$,

both with value 58/3, and the inverse symmetric division deg index does so at order 10 (exact enumeration in the repository).

3.3 Where the limit binds

By exhaustive enumeration of all trees of orders 2–17 in exact arithmetic (Table 2), two trees of order at most 12 with equal q-histograms have equal degree-pair profiles, and the same holds for trees with $\Delta \leq 4$ of order at most 15. With Theorem 3(iii), the pure-γ points at algebraic $\gamma \neq 0$ separate all distinct profiles exactly up to those orders and at no larger order. Among all trees, at orders 13–17, generic points off Π (Proposition 2) and the Randić index separate more trees than any point of Π. Degeneracy [23], measured as $100(1 - k/N)$ with k distinct values among N trees, is 25.5% for the pure-γ points on the 106 trees of order 10 (79 distinct values, the number of distinct profiles and so the floor for any degree-pair-based index), tied with R, GA and AG, while M_1 (83.0%) and M_2 (67.0%) are the most degenerate. GA can do worse than the pure-γ points: it merges two distinct q-histograms among trees first at order 17, for instance $\{(1,5)^{10},(2,2)^2,(5,5)^2,(2,5),(1,2)\}$ and $\{(4,5)^3,(1,4)^5,(1,5)^6,(2,5),(1,2)\}$.

Table 2. First tree order at which discrimination is lost, from all trees of orders 2 to 17. "Below floor": first order with fewer distinct values than distinct profiles (values/profiles). "Merges": first order at which an index on Π gives equal values to two trees with different q-histograms. Every count, ties included, is certified exactly: for the pure-γ points by Theorem 3(ii), and for R and GA by exact arithmetic in the radicals $(ij)^a(i+j)^b$ [6].

	all trees		trees with $\Delta \leq 4$	
Index	below floor	merges	below floor	merges
$LO(0,0,\gamma)$	13 (566/570)	never[a]	16 (809/810)	never[a]
R	14 (1095/1100)	n/a[b]	17 (1191/1194)	n/a[b]
GA	13 (566/570)	17	16 (809/810)	> 17[c]

[a] For every nonzero algebraic γ (Theorem 3(ii)). [b] R is not on Π. [c] No merge up to order 17, the largest examined.

Off Π the imbalance factor can restore discrimination lost at particular

rational exponents, and at rational exponents a nonzero algebraic γ never destroys any.

Proposition 4 (Parent distinctions survive)**.** *Let $\alpha, \beta \in \mathbb{Q}$ and let $\gamma \neq 0$ be algebraic. For graphs G, H and each imbalance value q write $c_q(G,H) = \sum_{(i,j)\colon q_{ij}=q} \big(m_{ij}(G) - m_{ij}(H)\big)(ij)^{\alpha}(i+j)^{\beta}$. Then $LO(G;\alpha,\beta,\gamma) = LO(H;\alpha,\beta,\gamma)$ if and only if $c_q(G,H) = 0$ for every q. In particular, if $M_{\alpha,\beta}(G) \neq M_{\alpha,\beta}(H)$ then $LO(G;\alpha,\beta,\gamma) \neq LO(H;\alpha,\beta,\gamma)$.*

Proof. The difference is $\sum_q c_q(G,H)\, e^{\gamma q}$. For rational α, β the coefficients $c_q(G,H)$ are algebraic, and the exponents γq are distinct algebraic numbers, so by the Lindemann–Weierstrass theorem [5] the sum vanishes only if every $c_q(G,H)$ vanishes. Finally $\sum_q c_q(G,H) = M_{\alpha,\beta}(G) - M_{\alpha,\beta}(H)$. ■

The algebraic hypothesis cannot be dropped: at $\gamma = \ln 4$, $LO(K_4;0,0,\gamma) = 6 = 3e^{\gamma/2} = LO(K_{1,3};0,0,\gamma)$. On all trees of order 16 (4069 profiles), $LO(0,\frac{1}{2},0)$ and $LO(0,\frac{1}{3},0)$ take only 3475 and 3476 distinct values, while $LO(0,\frac{1}{2},1)$ and $LO(0,\frac{1}{3},1)$ separate all 4069. These counts are exact (at $\gamma = 1$ by Proposition 4), but the effect is checked only up to order 16 and not proved for larger orders, and it yields no predictive gain over a tuned parent (Section 5). The sum-connectivity index $\chi = M_{0,-1/2}$ is not weakly discriminating over chemical trees [35, Example 2.2]; for the trees S, T used there, at $(\alpha,\beta) = (0,-\frac{1}{2})$ the difference $D(\gamma) = LO(S) - LO(T)$ satisfies $D(0) = 0$ and $D(1) \approx 0.2255$, but $D(10^{-6}) \approx 2.4 \times 10^{-7}$, so the separation is small for small γ.

4 Behaviour along the imbalance axis

Write $\theta = (\alpha,\beta,\gamma)$ and $z_{uv} = \big(\ln(d_u d_v), \ln(d_u + d_v), q_{uv}\big)$, so that $LO_G(\theta) = \sum_{uv} e^{\theta^{\top} z_{uv}}$ and $\log LO_G$ is a log-sum-exp function of θ.

Lemma 1 (Affine independence of the imbalance direction)**.** *For $\Delta \geq 3$ the augmented vectors $\big(1, \ln(ij), \ln(i+j), q_{ij}\big)$, $(i,j) \in P_\Delta$, span $\mathbb{R}^4$. Consequently no choice of (α',β') and scaling constant reproduces the action of $\gamma \neq 0$ on all admissible degree pairs. For $\Delta \leq 2$ it is reproduced exactly.*

Proof. On $(1,1),(2,2),(1,2)$ the vectors $(1,\ln(ij),\ln(i+j))$ have determinant $\ln 2\cdot\ln(9/8)\neq 0$, so it suffices that q_{ij} is not affine in $(\ln(ij),\ln(i+j))$. If $q_{ij}=a+b\ln(ij)+c\ln(i+j)$ on $(1,1),(2,2),(1,2),(1,3)$, the first two pairs give $a=-c\ln 2$ and $b=-c/2$, and the last two give $\frac{1}{3}=c\,(\ln 3-\frac{3}{2}\ln 2)$ and $\frac{1}{2}=c\,(\ln 2-\frac{1}{2}\ln 3)$, so $c\neq 0$ and $8\ln 3=13\ln 2$, i.e. $3^8=2^{13}$, which is false. Without the pair $(1,1)$, which occurs only in K_2, the vectors of $(1,2),(1,3),(2,2),(3,3)$ have determinant $-\frac{1}{6}\ln\frac{3}{2}\,\ln\!\big(2^{13}/3^8\big)\approx -0.0150\neq 0$. If $e^{\theta^\top z}=c\,e^{\theta'^\top z}$ on P_Δ, then $(\ln c,\theta'-\theta)$ is orthogonal to every augmented vector, hence zero, so $\theta=\theta'$. For $\Delta=2$ the three vectors $(1,\ln(ij),\ln(i+j))$ of $(1,1),(1,2),(2,2)$ already have rank 3, so q_{ij} is an affine function of $\ln(ij)$ and $\ln(i+j)$ on them and $e^{\gamma q_{ij}}$ equals a constant times $(ij)^a(i+j)^b$; for $\Delta=1$, $q\equiv 0$. ∎

Theorem 5 (Log-convexity)**.** *Let $F_G(\theta)=\log LO_G(\theta)$ and let $\mathbb{E}_\theta$, Cov_θ refer to the edge probability measure $w_{uv}(\theta)=e^{\theta^\top z_{uv}}/LO_G(\theta)$. Then $\nabla F_G=\mathbb{E}_\theta[z]$ and $\nabla^2 F_G=\mathrm{Cov}_\theta(z)\succeq 0$, so F_G is convex; it is strictly convex if and only if the z_{uv} do not lie in a proper affine subspace of $\mathbb{R}^3$. In particular $\log LO$ is non-decreasing and convex in γ, and strictly increasing unless every edge joins vertices of equal degree.*

Proof. Differentiation gives the gradient and Hessian, and $w(\theta)$ has full support, so $\mathrm{Cov}_\theta(z)$ is singular in a direction v if and only if $v^\top z_{uv}$ is constant over the edges. If no such v exists the Hessian is positive definite and F_G is strictly convex; if one exists, $t\mapsto F_G(\theta+tv)$ is affine and F_G is not strictly convex. Finally $\partial_\gamma F_G=\mathbb{E}_\theta[q]\geq 0$, with equality if and only if $q\equiv 0$. ∎

Proposition 6 (Order exchanges along γ)**.** *Fix (α,β) and graphs G, H, let $q_1<\cdots<q_r$ be the distinct imbalances of the degree pairs occurring in G or H, and write $D(\gamma)=LO(G;\alpha,\beta,\gamma)-LO(H;\alpha,\beta,\gamma)=\sum_{k=1}^{r}c_k e^{\gamma q_k}$ with $c_k=\sum_{q_{ij}=q_k}(m_{ij}(G)-m_{ij}(H))(ij)^\alpha(i+j)^\beta$. Let V be the number of sign changes in $(c_1,\dots,c_r)$ after zero terms are removed. If $D\not\equiv 0$, then D has at most $V\leq r-1$ real zeros counted with multiplicity, and their number has the parity of V. In particular, for graphs with $\Delta\leq 4$ the possible imbalances are $0,\frac{1}{7},\frac{1}{5},\frac{1}{3},\frac{1}{2},\frac{3}{5}$, and two such graphs exchange order at most five times as γ ranges over $\mathbb{R}$, for every (α,β).*

Proof. With $x = e^{\gamma}$, D is the generalised polynomial $\sum_k c_k x^{q_k}$, and Laguerre's extension of Descartes' rule of signs [22] bounds its positive zeros by V, with the parity of V. ∎

The bound is attained. The trees of order 12 with profiles $\{(1,3)^2, (1,4)^2, (1,6)^5, (3,4), (4,6)\}$ and $\{(1,2), (1,5)^4, (1,6)^4, (2,6), (5,6)\}$ have $V = 5$, and at $(\alpha, \beta) = (-1, 0)$ they cross at $\gamma \approx -17.2$, -12.0, -2.0, 2.7 and 31.1. Whether five crossings are attained for $\Delta \le 4$ is open.

For a graph G let $R(G)$ be the set of degree pairs it realises, $r(G)$ the number of distinct imbalances among them, and $A(G)$ the affine dimension of $\{z_{ij} : (i,j) \in R(G)\}$.

Proposition 7 (Per-graph identifiability)**.** *(i) $A(G)$ equals the rank of $\mathrm{Cov}_\theta(z)$ at every θ, and the directions v for which there is a constant c with $LO(G; \theta + tv) = e^{ct}\, LO(G; \theta)$ for all t and θ form a subspace of dimension $3 - A(G)$.*

(ii) $A(G) \le \min\big(|R(G)| - 1,\ r(G),\ 3\big)$.

(iii) If $r(G) = 2$ with imbalances $q_1 \ne q_2$, then $v = (1, -2, c)$ with $c = [\ln(1 - q_2^2) - \ln(1 - q_1^2)]/(q_1 - q_2)$ is such a direction.

Proof. (i) As in Theorem 5, $\mathrm{Cov}_\theta(z)$ is singular exactly in the directions v with $v^\top z$ constant on $R(G)$, a subspace of dimension $3 - A(G)$, and these are exactly the rescaling directions, since $v^\top \nabla F_G = c$ is constant if and only if $\mathrm{Cov}_\theta(z)\, v = 0$ for all θ. (ii) By (3), $z_{ij} = (2s + \ln(1 - q^2) - \ln 4,\ s,\ q)$ with $s = \ln(i + j)$, so pairs of equal imbalance lie on one line parallel to $(2, 1, 0)$, and r parallel lines lie in an affine space of dimension at most r. (iii) $v^\top z_{ij} = \ln(1 - q_{ij}^2) - \ln 4 + c\, q_{ij}$ takes the same value at q_1 and q_2. ∎

Equality holds in (ii) for every set of admissible pairs with $\Delta \le 6$ (by exhaustive enumeration), hence for all molecular graphs. On the octanes the union of realised pairs contains nine of the ten admissible pairs and has affine dimension 3, so the octane fits of Section 5 are free of exact confounding of this kind. On the spider with four legs of length three, which realises only the imbalances $\frac{1}{3}$ and 0, γ is confounded with (α, β) along the direction of part (iii).

Proposition 8 (Extremal trees of the pure-γ points)**.** *Let $n \geq 4$ and $\gamma > 0$. Among all trees of order n, the path P_n is the unique minimiser and the star S_n the unique maximiser of $LO(\cdot\,; 0, 0, \gamma)$, with*

$$(n-3) + 2e^{\gamma/3} \;\leq\; LO(T; 0, 0, \gamma) \;\leq\; (n-1)\, e^{\gamma(n-2)/n}.$$

For $\gamma < 0$ the roles of P_n and S_n are exchanged.

Proof. Let $\gamma > 0$. Every edge of a tree of order n has $q_{uv} \leq (n-2)/n$, with equality if and only if it is a $(1, n-1)$-edge, which gives the upper bound and S_n. A tree with $k \geq 2$ leaves has k edges incident to a leaf, each with $q \geq \frac{1}{3}$, since the other end has degree $d \geq 2$ and $(d-1)/(d+1)$ is increasing in d. Hence $LO(T; 0, 0, \gamma) \geq (n-1-k) + ke^{\gamma/3} \geq (n-3) + 2e^{\gamma/3}$, since $e^{\gamma/3} > 1$; equality forces $k = 2$, that is $T = P_n$, which attains the bound with two $(1, 2)$-edges and $n-3$ $(2, 2)$-edges. For $\gamma < 0$ every inequality reverses. ∎

Proposition 8 is a consequence of the criteria of Cruz and Rada [9]; the direct proof above is included because it is short. For the weights $\varphi_{ij} = e^{\gamma q_{ij}}$ and $E = e^{\gamma/3}$, their function $f(i, j) = \frac{ij}{i+j}(\varphi_{ij} + 2\varphi_{12} - 3\varphi_{22})$ equals $2(E-1)$ at $(1, 2)$ and $(2, 2)$. For $\gamma > 0$ it is larger at every other pair: if $i, j \geq 2$ then $\varphi_{ij} + 2E - 3 \geq 2(E-1) > 0$ and $ij/(i+j) > 1$, and for $(1, j)$, $j \geq 3$, one has $f \geq \frac{3}{4}(e^{\gamma/2} + 2E - 3) > 2(E-1)$, since with $x = e^{\gamma/6} > 1$ this reads $3x^3 - 2x^2 - 1 = (x-1)(3x^2 + x + 1) > 0$. Theorem 2.1 of [9] then gives the lower bound with equality only for P_n, and their Theorem 3.1 gives the upper bound with equality only for S_n, because $e^{\gamma q}$ is largest at the pair $(1, n-1)$. For $\gamma < 0$ every inequality reverses. Among chemical trees the bound $LO(T; 0, 0, \gamma) \leq (n-1)e^{3\gamma/5}$ for $\gamma > 0$ is attained only for $n = 5$, and the maximal chemical trees for $n > 5$ are left open.

5 QSPR diagnostics

We compare the family with the classical indices it reduces to on the 18 octane isomers, following the benchmark of the diminished Sombor

study [29], and use the 35 nonane isomers as a replication set. No parameter in the correlation comparison is tuned against the property data. The repository history begins with the octane correlations already computed, so it cannot document that the benchmark points $\gamma = 1$ and 2 were fixed before the octane data were seen. The sets are small, so the octane correlations are descriptive. After the nonane result of Section 5.2 we made the pre-registered tests on measured boiling points of Section 5.3 the primary predictive evidence.

5.1 Correlations on the octane isomers

Table 3 reports Pearson correlations with five properties. The property values are the standard benchmark compilation, audited against the NIST WebBook [26] and the API Research Project 44 tables [39]. Seven entropies and the compiled acentric factors rest on secondary sources, and the entropies can move the 0.001 margin between HM and M_1, so the ordering of the S column is not interpreted. The vaporisation enthalpy of 2,2,3,3-tetramethylbutane, which is crystalline at 298 K, has no stated source; with the isomer excluded $LO(0, 0, 1)$ stays first, level at three decimals with $LO(0, 0, 2)$ ($|r| = 0.980$), and with the hypothetical-liquid estimate it gives $|r| = 0.976$, while the NIST sublimation enthalpy lowers every $|r|$ to 0.374 or below, so this column is conditional on a liquid-phase value. Deleting single isomers shifts each bolded $|r|$ by at most 0.027. The descriptors are nearly collinear: fourteen pairs have $|r| \geq 0.99$, the first two principal components carry 99.05% of the variance, and $LO(0, 0, 2)$ has $|r| = 0.993$ with a classical reduction. Differences in the third decimal are therefore not rankings, and the bolded entries, each the largest of twelve correlations on 18 molecules, are optimistic. The two collision pairs of Table 1 differ by 1.2 °C and 2.1 °C in T_B although every vertex-degree-based index gives them equal values.

5.2 Freeing γ and the search box

We ask whether, at identical search budgets, freeing γ improves out-of-sample prediction over an optimised parent $M_{\alpha,\beta}$, in a fully nested

Table 3. Pearson correlation r between the Loyola reductions, the fixed pure-γ benchmark points and the five physicochemical properties of the 18 octane isomers. Bold flags the largest $|r|$ per column, and this is descriptive. Margins to the runner-up are 0.001–0.032, and their paired bootstrap intervals include zero (Supplementary Section S1). These intervals are post-selection and not adjusted for multiplicity. The entry -0.000 (ISI, ΔH_f) is -0.0002.

Index	T_B	ΔH_f	ΔH_{vap}	S	ω
$M_1 = LO(0,1,0)$	-0.718	-0.762	-0.935	-0.970	-0.970
$M_2 = LO(1,0,0)$	-0.497	-0.542	-0.811	-0.934	$\mathbf{-0.988}$
$HM = LO(0,2,0)$	-0.654	-0.710	-0.903	$\mathbf{-0.971}$	-0.982
${}^{m}M_2 = LO(-1,0,0)$	$\mathbf{+0.855}$	$\mathbf{+0.891}$	$+0.925$	$+0.863$	$+0.794$
$R = LO(-1/2,0,0)$	$+0.819$	$+0.850$	$+0.958$	$+0.932$	$+0.898$
$\chi = LO(0,-1/2,0)$	$+0.800$	$+0.832$	$+0.961$	$+0.947$	$+0.924$
$H/2 = LO(0,-1,0)$	$+0.821$	$+0.849$	$+0.961$	$+0.932$	$+0.900$
$ISI = LO(1,-1,0)$	-0.005	-0.000	-0.383	-0.593	-0.738
$GA = 2\,LO(1/2,-1,0)$	$+0.822$	$+0.858$	$+0.965$	$+0.939$	$+0.906$
$AG = \frac{1}{2}LO(-1/2,1,0)$	-0.816	-0.859	-0.957	-0.938	-0.900
$LO(0,0,1)$	-0.828	-0.832	$\mathbf{-0.984}$	-0.928	-0.929
$LO(0,0,2)$	-0.829	-0.846	-0.981	-0.937	-0.925

leave-one-out design with random search (Supplementary Section S3); a parameter-optimised parent has been fitted to octane properties before [3]. In the box $(-2, 2)$ freeing γ improved the nested Q^2 for octane ΔH_{vap} in 100/100 candidate streams at budgets 200 and 500, with median $\Delta Q^2 = +0.069$ and $+0.068$, and in 93/100 and 96/100 with 2,2,3,3-tetramethylbutane excluded. It usually degraded ΔH_f and was mixed elsewhere. For ΔH_{vap}, however, the parent chose $(\alpha, \beta) = (1.021, -1.993)$, on the boundary of the box, in 16 of the 18 folds. By (3) the parent reaches imbalance information through $(1 - q^2)^{\alpha}$ along $\beta \approx -2\alpha$, which needs large exponents and is the direction the box truncates. A box of half-width 12, chosen after inspecting the parent's nested Q^2 curves on the octanes and the nonanes as the smallest examined in which its selections are interior, was used below for every property. It does not favour the parent uniformly: for T_B it lowers the parent's nested Q^2 from 0.665 to 0.606 on the octanes.

The matched comparison was repeated without pre-registration, with 2000 candidates per model and seeds 0–49. In the wide box the

vaporisation-enthalpy gain shrinks to about zero: a hybrid control that widens only (α, β) and keeps $\gamma \in (-2, 2)$ gives median $\Delta Q^2 = +0.004$ on the octanes (28 of 50 seeds) and -0.002 on the nonanes (24 of 50), while widening all three coordinates gives -0.010 and -0.016. Most of the change comes from the parent improving in the wide box ($0.887 \to 0.942$ on the octanes). Octane S moves from 11 to 38 wins, which shows how strongly the win count depends on the search protocol. Equal budgets do not give equally thorough searches in two and three dimensions, so the design may cost the three-parameter model part of the effect sought.

Under a protocol committed before the nonane data were committed and before any model was fitted to them (commit `964b127`), the narrow-box design was applied to the 35 nonane isomers. The 35 boiling points and 34 standard vaporisation enthalpies are from the NIST WebBook [26]; 29 of the enthalpies are compiled values, 4 listed under Labbauf et al. [24] and 25 under Reid [37], a review of the handbook [47], so agreement with them measures consistency with compiled values. The pre-registered verdict is that the γ effect *replicates* for ΔH_{vap}, with LO better than GM in 95/100 and 98/100 seeds and median $\Delta Q^2 = +0.048$ and $+0.053$, and is *mixed* for T_B (58/100 and 63/100). We report the verdict as obtained. Its seed-count rule counts optimiser variance on fixed data, which we regard as a design error. A post hoc molecule bootstrap conditional on the fitted predictions gives a 95% interval for LO minus GM on nonane ΔH_{vap} of $[-0.006, +0.113]$ at budget 200, which includes zero, and in the wide box, which is not pre-registered, the gain shrinks to about zero. The untuned point $LO(0, 0, 1)$, which correlates at 0.996 with $IRLA$ on both sets, matches or beats the tuned Loyola model in point estimate (0.957 and 0.903 against medians 0.953 and 0.900; on the octanes this point was singled out from the same data). Ridge regression on the degree-pair counts has a higher point estimate than every index model on the nonanes ($Q^2 = 0.927$ for ΔH_{vap}, 0.682 for T_B; the conditional interval for LO minus ridge on nonane ΔH_{vap}, $[-0.101, +0.032]$, includes zero), but on the octanes it is below the best single index for four of the five properties and level on ΔH_{vap} (0.954 against 0.957; Supplementary Table S7).

5.3 Measured boiling points

Two tests used measured boiling points from the NIST WebBook [26], extracted by a rule committed to the repository (commit `a5cf378`) before any model was fitted. The rule admits all hexanes, heptanes and octanes, 34 of the 35 nonanes and 34 of the 75 decanes. The included and excluded decanes do not differ detectably in branching or imbalance (Supplementary Section S9), but with 34 and 41 molecules this does not show that the decane cohort is representative. Test A refits and reselects inside each leave-one-out fold on the decanes. Test B pools all 100 molecules with the carbon number n_C as a covariate; the octane and nonane boiling points were used above, so Test B is a pooled size-adjusted analysis rather than a fresh external test. Both use the wide boxes, $(\alpha, \beta) \in (-12, 12)^2$ and $\gamma \in (-2, 2)$, with 2000 candidates and seeds 0–49. The pre-registered criterion for a γ effect required at least 40 wins in 50 seeds and a molecule-bootstrap interval excluding zero, which we read as requiring the lower end of the two-sided 95% interval to exceed zero.

Table 4. Measured boiling points (nested leave-one-out Q^2). Test A: decane cohort, refitted within folds. Test B: pooled C6–C10 with the carbon number n_C in every index model (Class., $IRLA$, LO_1: n_C plus that descriptor); Ridge uses the counts alone, whose sum is $n_C - 1$. Base: training-fold mean (Test A) or n_C alone (Test B). Class.: best classical index chosen in each fold. LO_1: fixed $LO(0, 0, 1)$. Ridge: ridge regression on the degree-pair counts. GM, LO_h: medians over seeds 0–49 in the wide boxes, LO_h being the hybrid control with (α, β) in the wide box and $\gamma \in (-2, 2)$. ΔQ^2: median over seeds of the paired difference LO_h minus GM. The registered intervals are conditional on the fitted predictions and omit refitting and selection uncertainty (see below). Last column: seeds (of 50) in which LO_h beats GM.

Test	Base	Class.	$IRLA$	LO_1	Ridge	GM	LO_h	ΔQ^2	>
decanes ($n = 34$)	−0.062	0.176	0.167	0.149	0.860	0.514	0.406	−0.072	15
C6–C10, with n_C ($n = 100$)	0.938	0.967	0.963	0.963	0.991	0.977	0.977	−0.0001	25

Both pre-registered verdicts are negative. On the decanes the median $\Delta Q^2 = -0.072$, LO is ahead in 15 of 50 seeds, and the 95% interval is $[-0.156, +0.001]$; on the pooled set the values are -0.0001, 25 of 50 and $[-0.003, +0.002]$. In the original box of half-width 2, reported as the

pre-registration requires, the two tuned models are level on the decanes (medians 0.469 and 0.465), and the Loyola model is marginally ahead on the pooled set (0.980 against 0.976); seed-matched comparisons of that narrow-box Loyola model with the parent were not registered, carry no interval and count optimiser variance, so they do not indicate a gain from γ. The registered intervals are conditional on the fitted predictions and omit refitting and selection. An exploratory resampling that reruns the whole nested procedure on random 80% subsamples gives, on the decanes (500 subsamples), median -0.110 with central 95% range $[-0.510, +0.312]$, and on the pooled set (200 subsamples) $+0.001$ with range $[-0.015, +0.008]$; these ranges are not confidence intervals for the full-sample ΔQ^2. Each replicate uses a single candidate stream, so the ranges also contain optimiser variance. No smallest effect of interest or equivalence margin was registered. The negative verdicts therefore mean that no gain was demonstrated under these protocols; they do not establish equivalence, the decane range neither supports nor excludes gains of practical size, and they do not show that γ never helps. On the registered secondary question of Test B, $IRLA$ and $LO(0, 0, 1)$ do not beat size plus the best classical index (0.963 against 0.967), while both tuned models have higher point estimates (0.977; no interval was registered for this secondary question).

Ridge regression on the ten degree-pair counts reaches $Q^2 = 0.860$ on the decanes, where every single index stays between 0.149 and 0.514 (medians over seeds for the tuned models), and 0.991 on the pooled set. A permutation check that repeats the whole nested procedure, including the choice of the ridge penalty, for 200 permutations of the decane boiling points never reaches 0.860 (median -0.09, maximum 0.20; empirical $p = 1/201 \approx 0.005$). Wiener's two-descriptor model [46], with the path number W and the number p_3 of carbon pairs three bonds apart, fitted without tuning, gives $Q^2 = 0.904$ on the decanes. On trees $p_3 = \sum_{uv}(d_u - 1)(d_v - 1)$ is a linear function of the degree-pair counts, so the distance information in this model is W. This single point estimate, given without an interval and for descriptors historically chosen on alkane boiling points, suggests that within one carbon number W may add to the degree-pair information; its difference from ridge is not resolved here. Under the stated boxes

and candidate budgets, ridge regression predicts the boiling points more accurately than every single-index model evaluated, with or without γ, in point estimate; the margin is large on the decanes and small on the pooled set, where size alone gives $Q^2 = 0.938$, and no interval for these differences is reported here. These conclusions concern acyclic alkanes.

6 Discussion and conclusion

Every index that depends on an edge only through its degree ratio sees a graph only through its histogram of normalised imbalances. For trees this histogram carries the full degree-pair information up to order twelve, and loses it at order thirteen, and at sixteen for chemical trees, with explicit colliding families. The pure-imbalance points of the Loyola family at nonzero algebraic γ attain this limit exactly; along the imbalance axis two chemical graphs exchange order at most five times; the extremal trees of the pure-imbalance points are the path and the star; and at rational exponents a nonzero algebraic γ never destroys a distinction the parent makes. The minimality of the collision orders, equality in Proposition 7(ii) for $\Delta \leq 6$ and the recovery of discrimination off Π rest on exhaustive enumeration; no general argument is given.

On alkanes no predictive gain from γ was demonstrated under the stated protocols, which may handicap the three-parameter search, and on measured boiling points ridge regression on the degree-pair counts had higher point estimates than every single index evaluated. The data sets are small and acyclic: 18 octanes, 35 nonanes, 34 decanes and 100 pooled C6–C10 alkanes. Seven octane entropies and the compiled acentric factors rest on secondary sources, and the vaporisation enthalpy of 2,2,3,3-tetramethylbutane has an unresolved reference state. Changes made after outcomes had been seen are listed in Table 5; none changes a pre-registered verdict as reported, but the wide-box analysis that reverses the nonane conclusion is itself not pre-registered. Open problems are whether five crossings occur for $\Delta \leq 4$, the maximal chemical trees of the pure-imbalance points, and the behaviour on other compound classes.

Data availability. Data, scripts and verifiers that recompute every tab-

Table 5. Changes to the pre-registered protocols made after outcomes had been seen. Protocols and change records are in the reproducibility package [30].

Test	Change after outcomes	Effect
Nonanes (Section 5.2)	Parent box widened from half-width 2 to 12, chosen after inspecting the ΔH_{vap} curves	Registered verdict reported as obtained; the gain shrinks to about zero in the wide box
Nonanes	Seed-count criterion judged to count optimiser variance	None on reported numbers
Nonanes	Measured boiling points made the primary predictive evidence	None on reported numbers
Boiling points, Test B	All NIST pages of a skeleton pooled; one nonane excluded, one heptane value moved by 0.2 °C	Verdict unchanged; both cohorts reported
Nonanes	A second NIST T_B of 3-ethyl-4-methylhexane, 6.4 K lower, found later	Still mixed if excluded or replaced (Suppl. S4)
Boiling points, Test B	Least-squares solver rescaled by columns	Verdict unchanged
Boiling points, Test B	Scale-invariant rank test	No displayed value changes
Boiling points, Test B	Stricter rank threshold and a deletion-rank guard in the size-adjusted scorer	No output changes
Boiling points, both	Interval rule read as one-sided	None
Boiling points, both	Selection-aware resampling added (Section 5.3)	Exploratory; no verdict depends on it

ulated value are in the reproducibility package [30], where the nonane and boiling-point protocols were committed before the corresponding models were fitted. The NIST WebBook pages from which the experimental values were extracted are not redistributed; a script of the package downloads them and checks each against the copy from which the data sets were built. Lean 4 with Mathlib checks the convexity of the log-sum-exp form of $\log LO$, the spanning statement of the independence lemma and the order-thirteen profile collision on the pure-γ line; the transcendence, Laguerre and enumeration arguments are not machine-checked.

***Acknowledgment*:** The authors thank colleagues for helpful discussions.

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